The effect of bilberry nutritional supplementation on night visual acuity and contrast sensitivity.

PURPOSE: The purpose of this study was to investigate the effect of bilberry on night visual acuity (VA) and night contrast sensitivity (CS). METHODS: This study utilized a double-blind, placebo-controlled, crossover design. The subjects were young males with good vision; eight received placebo and seven received active capsules for three weeks. Active capsules contained 160 mg of bilberry extract (25-percent anthocyanosides), and the placebo capsules contained only inactive ingredients. Subjects ingested one active or placebo capsule three times daily for 21 days. After the three-week treatment period, a one-month washout period was employed to allow any effect of bilberry on night vision to dissipate. In the second three-week treatment period, the eight subjects who first received placebo were given active capsules, and the seven who first received active capsules were given placebo. Night VA and night CS was tested throughout the three-month experiment. RESULTS: There was no difference in night VA during any of the measurement periods when examining the average night VA or the last night VA measurement during active and placebo treatments. In addition, there was no difference in night CS during any of the measurement periods when examining the average night CS or the last night CS measurement during active and placebo treatments. CONCLUSION: The current study failed to find an effect of bilberry on night VA or night CS for a high dose of bilberry taken for a significant duration. Hence, the current study casts doubt on the proposition that bilberry supplementation, in the forms currently available and in the doses recommended, is an effective treatment for the improvement of night vision in this population.